INSTITUTE OF TECHNOLOGY, NIRMA UNIVERSITY

A Real-Time Two-Wheeler Violation Detection System Using Raspberry Pi and AI-Based Computer Vision

25MCA035 Sneh Patel 25MCA037 Milan Poria

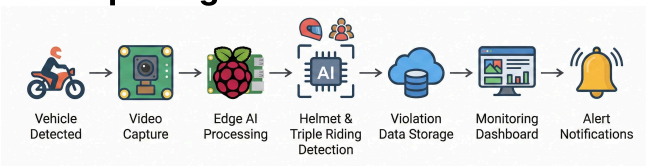
Problem Introduction

Helmet less riding and triple riding are common traffic violations on university campuses. These violations increase the risk of road accidents and compromise rider safety. Manual monitoring by security personnel is inefficient and cannot continuously observe all traffic activity. Therefore, automated systems using computer vision and IoT technologies are being developed to detect violations in real time and improve traffic safety monitoring [1], [3].

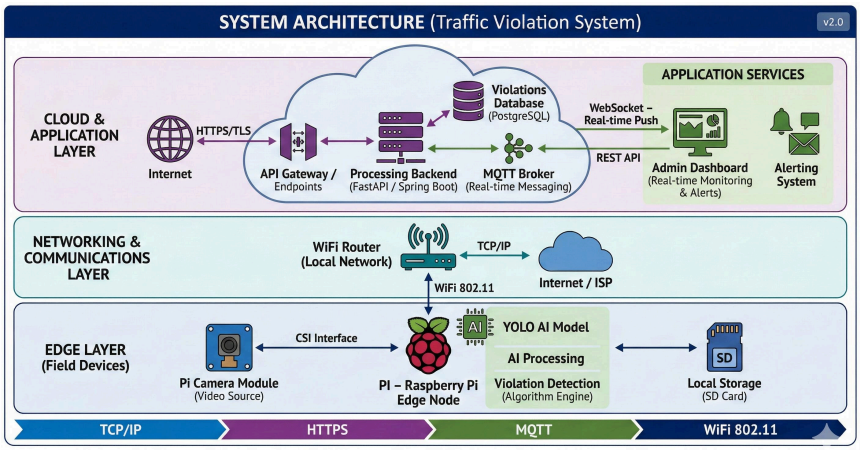
Proposed Solution

The proposed system is an IoT-based intelligent traffic monitoring solution that uses a Raspberry Pi and AI-based computer vision model to detect two-wheeler violations such as helmet less riding and triple riding. A Pi Camera Module captures real-time video and sends it to the Raspberry Pi for edge processing. The system applies an AI object detection model to identify violations and records the event. The detected violations are then transmitted to a cloud server where they can be monitored through an admin dashboard with alert notifications [2], [4].

Concept Diagram

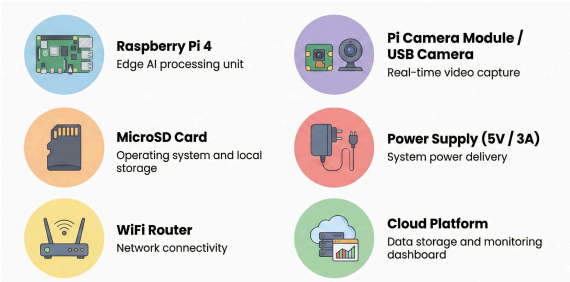


Conceptual representation of the IoT-based two-wheeler violation monitoring system.



System Architecture

Components Used



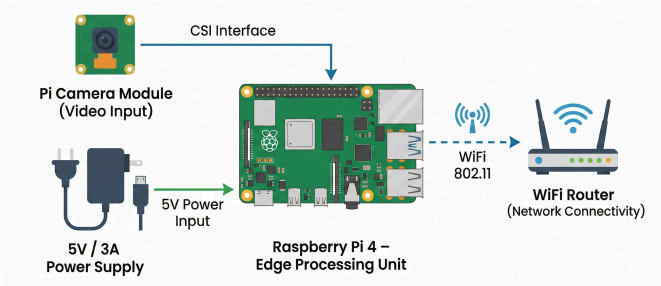
Brief Working of the System

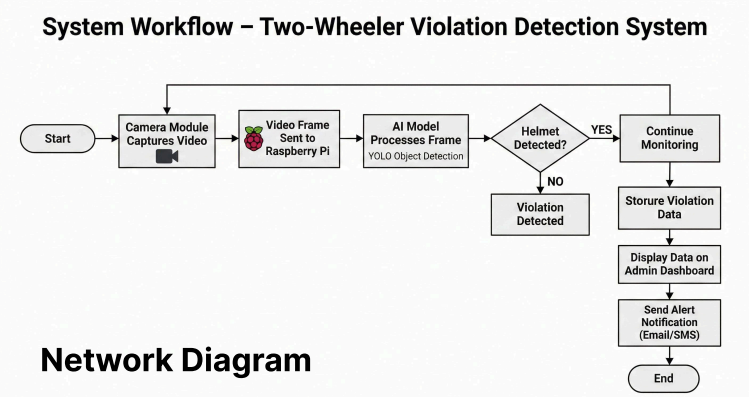
- The Pi Camera Module captures real-time video from the monitoring location [2].
- The Raspberry Pi processes video frames using an AI-based object detection model [2], [4].
- The detection model identifies helmet usage and counts the number of riders [1], [3].
- If a violation is detected, the system captures an image of the event [1].
- Violation data is stored locally and transmitted to the cloud server for monitoring [2], [4].
- The admin dashboard displays violation information for authorities [4].
- Alert notifications are generated to inform monitoring personnel [2].

Main Functionality

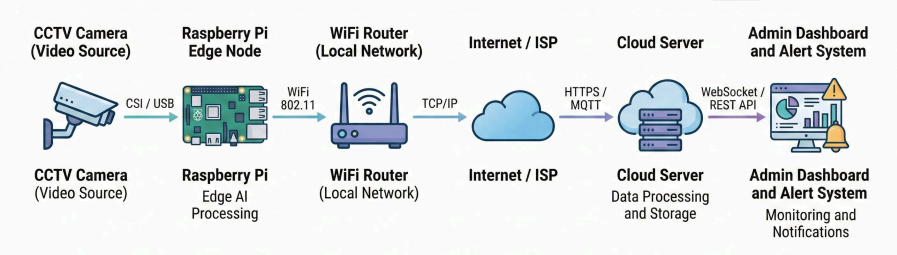
- Real-time helmet violation detection [1], [3]
- Triple riding detection using AI-based vision models [1]
- Edge AI processing using Raspberry Pi [2], [4]
- Automatic violation image capture [1]
- Cloud-based violation data storage [2], [4]
- Real-time monitoring dashboard [4]
- Alert notifications for authorities [2]

Circuit Diagram





Network Diagram



Benefits

- Improves campus road safety [3]
- Enables automated traffic monitoring [1], [2]
- Reduces manual supervision requirements [3]
- Provides a low-cost IoT-based monitoring solution [2]
- Scalable for smart city traffic monitoring systems [3]

Limitations

- Detection accuracy depends on lighting conditions and camera quality [1]
- Raspberry Pi has limited computational capability for complex AI models [2]
- Camera placement affects detection accuracy [3]